Week 3:
Data Structures

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Data Structures

Definition

refers to logical collection of data, collection of data that defines the structures being used.

Show relationships among the data items using primitive structures or other data structures available in a programming environment to facilitate operations of the ADTs.

Example**:**

records
files
arrays

Data Structures are programmed using features of programing languages. Simple data structures are programmed using **primitive data types** while complex data structures use **composite or user-defined data types**.

DATA TYPE

is the term used to refer to the kinds of data that a variable may “hold” in a given programming language.

DATA OR MATERIALS

INFORMATION
Processed data

DATA ITEM/FIELD
Smallest unit of information

Example:
EMPLOYEE’S NAME
A single elementary unit

TYPES OF DATA ITEM

GROUP ITEM

Item that can be subdivided

Example: NAME (FN MI LN)

ELEMENTARY ITEM

Could not be divided into sub items

Example: SSS_No , Employee_No

TERMS

RECORDS

Is a collection of related data items or fields.

FILE

Is a collection of related records

ENTITY

Is something that has certain properties

ATTRIBUTE

Is a property of an entity

DATA TYPE

Is a set of values that the variable may assume

CLASSIFICATION OF DATA STRUCTURE

LINEAR

It is elements from a sequence

NON-LINEAR

Trees/graphics

More questions?

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